

APPM2360 Project 1: How Much Does that House Really Cost?

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February 17, 2026

Contents

1	Introduction	1
2	Analysis of Fixed Rate Mortgages	2
2.1	Continuous Compounding	2
2.2	Behavior of the Loan Value	2
2.3	Solution for Principal	3
2.4	Monthly Payments	3
2.5	Interest Paid	3
2.6	Down Payment	4
2.7	30 vs 10 Year Fixed Rate	5
3	Numerical Solutions	5
3.1	Fixed Rate Mortgage	5
3.1.1		5
3.1.2		5
3.1.3		6
3.2	Adjustable Rate Mortgage (ARM)	6
3.2.1		6
3.2.2		6
3.2.3		7
3.2.4		7
4	Conclusion	7
5	Appendix	8
.1	Solving Eq. (3) for A_0 , r , and p	8
.2	MATLAB Code	8

List of Figures

2.1	Loan Value (Cost of Loan Over Time)	2
3.1	Euler vs True ($h=0.5$)	5
3.2	Euler vs True ($h=0.01$)	6
3.3	ARM ($h=0.01$)	7

1 Introduction

For this study, the focus is to analyze and compare mortgage options for prospective home buyers in Boulder, Colorado. Varying models and rates to project and study long term implications are examined. By framing each model through different equations, the optimal financial path is able to be deduced.

2 Analysis of Fixed Rate Mortgages

2.1 Continuous Compounding

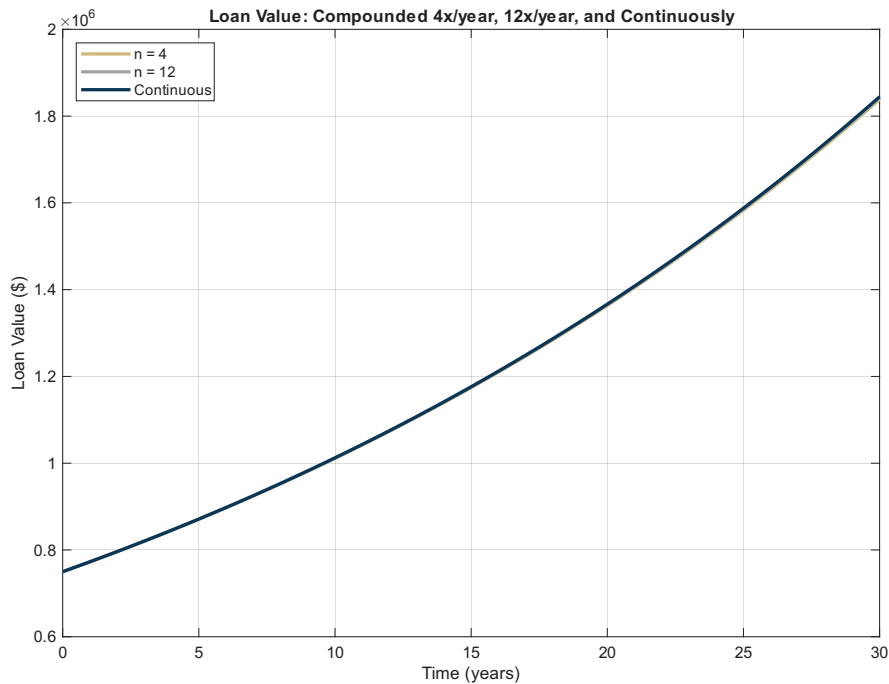


Figure 2.1: Loan Value (Cost of Loan Over Time)

Interest is calculated when a loan compounds. This frequency can affect the cost of the loan over time. This graph shows the effect of continuous compounding on the value of a \$750,000 loan, with an interest rate of 3% over a course of thirty years.

Compounding	Loan Cost
n=1	\$869455.56
n=2	\$870405.62
n=4	\$870888.11
n=12	\$871212.59
Continuous	\$871375.68

Table 1: Total Loan Cost After 5 Years

The total difference between a loan compounding once a year and compounding continuously is just under two thousand dollars for a thirty year mortgage. In the entire time paying for the loan this is a relatively small difference.

2.2 Behavior of the Loan Value

Finding an equilibrium solution for equation three can aid in the understanding of the behavior of the principal of the loan.

$$\frac{dA}{dt} = rA - 12p$$

To find the equilibrium of equation 3, set $\frac{dA}{dt} = 0$ to find the equilibrium value A_{eq} :

$$\begin{aligned} 0 &= rA_{eq} - 12p \\ rA_{eq} &= 12p \\ A_{eq} &= \frac{12p}{r} \end{aligned}$$

If $A > A_{eq}$, then $rA > 12p$ so $\frac{dA}{dt} > 0$ (loan grows). If $A < A_{eq}$, then $rA < 12p$ so $\frac{dA}{dt} < 0$ (loan shrinks). Based on this, the equilibrium is unstable.

2.3 Solution for Principal

To find the exact behavior of the loan in this specific situation, the differential equation with an arbitrary r and p can be solved. The derivation found in appendix .1.

$$A(t) = \frac{12p}{r} + \left(A_0 - \frac{12p}{r} \right) e^{rt}$$

2.4 Monthly Payments

To find the monthly payment p that pays off the loan in t_l years, set $A(t_l) = 0$ in the solution.

Set $A(t_l) = 0$ and solve for p :

$$0 = \frac{12p}{r} + \left(A_0 - \frac{12p}{r} \right) e^{rt_l}$$

$$p = \frac{A_0 r e^{rt_l}}{12(e^{rt_l} - 1)}$$

Plugging in the numbers:

For the 10-year, 3% mortgage:

$$p = \frac{750,000 \times 0.03 \times e^{0.03 \times 10}}{12(e^{0.03 \times 10} - 1)} = \$7,234.30$$

For the 30-year, 5% mortgage:

$$p = \frac{750,000 \times 0.05 \times e^{0.05 \times 30}}{12(e^{0.05 \times 30} - 1)} = \$4,022.55$$

It is shown that over 30 years with a 5% mortgage, the monthly cost is drastically decreased compared to the 10 year with a 3% mortgage. The monthly payment for the 30 year mortgage is almost half of that of the 10 year mortgage.

2.5 Interest Paid

The total cost of the loan is determined by calculating the sum of every monthly payment over the entire duration of the agreement. This approach allows for the calculation of the exact interest paid by subtracting the initial principal from the cumulative total. These metrics provide a clear comparison between a 10-year mortgage at a 3 percent interest rate and a 30-year mortgage at a 5 percent interest rate, highlighting the long-term financial implications of each structure.

Total Interest Paid

The total amount paid is:

$$\text{Total Paid} = 12p \times t_l$$

The total interest paid is:

$$\text{Interest Paid} = \text{Total Paid} - A_0$$

Plugging in the numbers:

For the 10-year, 3% mortgage:

$$p = 7,234.30$$

$$t_l = 10$$

$$A_0 = 750,000$$

$$\text{Total Paid} = 12 \times 7,234.30 \times 10 = \$868,116.00$$

$$\text{Interest Paid} = 868,116.00 - 750,000 = \$118,116.00$$

For the 30-year, 5% mortgage:

$$p = 4,022.55$$

$$t_l = 30$$

$$A_0 = 750,000$$

$$\text{Total Paid} = 12 \times 4,022.55 \times 30 = \$1,447,118.00$$

$$\text{Interest Paid} = 1,447,118.00 - 750,000 = \$697,118.00$$

It is seen that the longer the loan duration is, the greater the interest amount will be as after 30 years. In this case the total interest paid is 5.75 times that of what is paid after 10 years.

2.6 Down Payment

Making a down payment is a strategy often used to lower the total amount of money borrowed from the bank. By paying a portion of the house's cost upfront, the starting size of the loan is reduced, which can lead to significant savings over time. If the down payment was \$100,000 for a \$750,000 house, the initial loan amount would only be \$650,000. The lower starting balance means that the monthly charge would be greatly decreased starting at the beginning of the mortgage. The following analysis compare the 10 year and 30 year mortgages.

Effect of a Down Payment

If 100,000 is paid up front, the principal becomes $A_0 = 650,000$.

Calculate new monthly payments:

For the 10-year, 3% mortgage:

$$p = \frac{650,000 \times 0.03 \times e^{0.03 \times 10}}{12(e^{0.03 \times 10} - 1)} = \$6,273.06$$

For the 30-year, 5% mortgage:

$$p = \frac{650,000 \times 0.05 \times e^{0.05 \times 30}}{12(e^{0.05 \times 30} - 1)} = \$3,486.21$$

Calculate new total and interest paid:

For the 10-year, 3% mortgage:

$$\text{Total Paid} = 12 \times 6,273.06 \times 10 = \$752,747.20$$

$$\text{Interest Paid} = 752,747.20 - 650,000 = \$102,747.20$$

For the 30-year, 5% mortgage:

$$\text{Total Paid} = 12 \times 3,486.21 \times 30 = \$1,255,035.60$$

$$\text{Interest Paid} = 1,255,035.60 - 650,000 = \$605,035.60$$

Savings:

For the 10-year, 3% mortgage:

$$\text{Interest Saved} = 118,116.00 - 102,747.20 = \$15,368.80$$

For the 30-year, 5% mortgage:

$$\text{Interest Saved} = 697,118.00 - 605,035.60 = \$92,082.40$$

It is seen that in the 10 year mortgage with an \$100,000 down payment, \$14,368.80 is saved over the 10 year mortgage without the down payment. Additionally, with a 30 year mortgage with an \$100,000 down payment, a massive \$92,082.40 is saved. Through these two simply analysis, it can be seen how much is saved by simply using a down payment before obtaining a loan.

2.7 30 vs 10 Year Fixed Rate

An advantage of a 30 year fixed rate mortgage over a 10 year fixed rate mortgage is that the monthly payments will be much lower for a 30 year. However the total cost of the loan will be much greater in said 30 year mortgage.

3 Numerical Solutions

3.1 Fixed Rate Mortgage

In the following analysis, a mortgage of \$750,000 with a constant interest rate of %5 and a monthly payment of \$4,000 along is considered along with a second scenario with the same bounds however the monthly payment is \$4,500 is also considered.

3.1.1

When Euler's method with a step size of 0.5 is used, an approximation is achieved that the loan will be paid off in 31 years. However, if Euler's method with a step size of 0.01 is used, a much higher accuracy approximation shows that the loan will be paid off in 30.41 years.

3.1.2

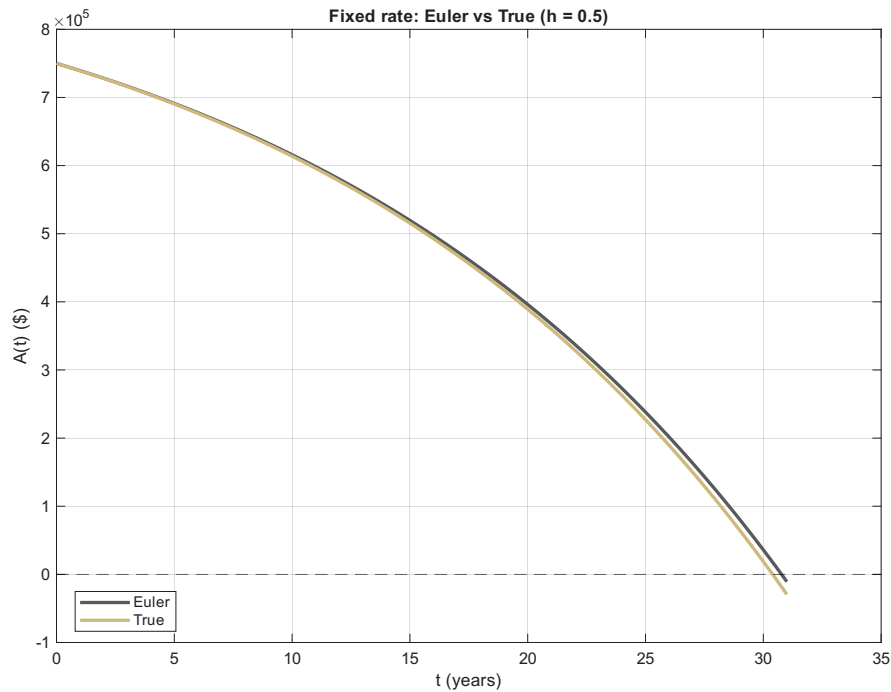


Figure 3.1: Euler vs True (h=0.5)

The approximation using Euler's method with a step size of h=0.5 is plotted next to the true solution. The true solution suggests that the mortgage is paid off with just six months less than the Euler

solution. Euler is a good approximation for this case with this step size, however not perfect. This lack of perfection is due to the error achieved using Euler's method.

3.1.3

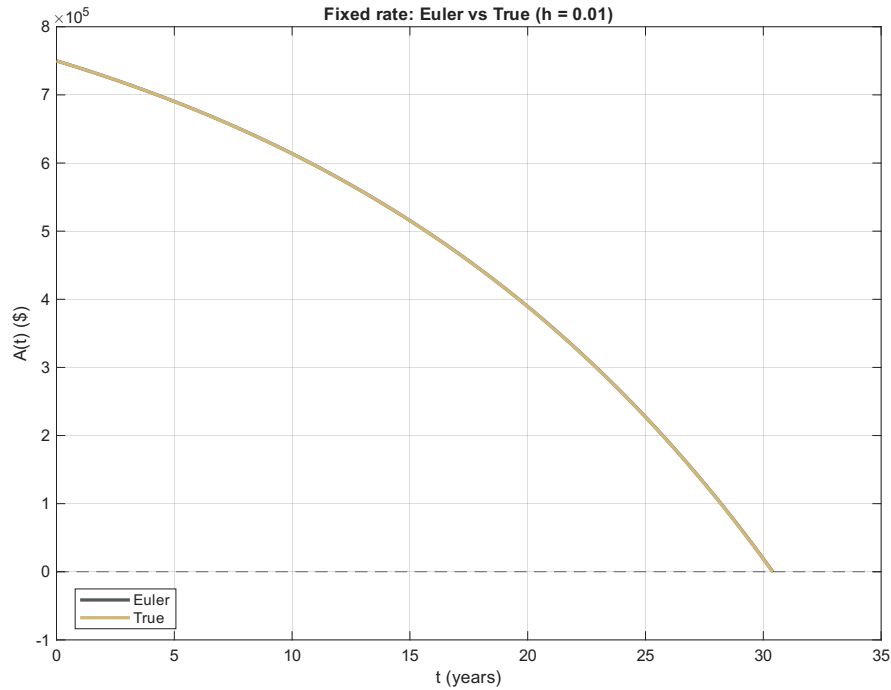


Figure 3.2: Euler vs True (h=0.01)

The approximation using Euler's method with a step size of $h=0.01$ is plotted next to the true solution. The Euler approximation is very accurate to the true solution. Euler is a great approximation for this case with this step size as a lower step size decreases the error gained through using Euler's method.

3.2 Adjustable Rate Mortgage (ARM)

Transitioning to adjustable-rate mortgages, the analysis considers a 750,000 dollars loan that begins with a lower fixed interest rate of 3 for the initial five years. After those first five years, the bank changes the rate based on what is happening in the credit markets. This change is calculated using

$$r(t) = 0.03 + 0.015\sqrt{t - 5} \quad (1)$$

where the interest rate goes up over time. By using Euler's method with a small step value of $h = 0.01$, the following analysis shows how much extra interest is paid and how long it takes to own the home based on different monthly payments.

3.2.1

By using Euler's method of a fixed rate mortgage using an h value of 0.01 and while paying \$4000 per month, it will take 44.90 years to pay off the mortgage.

3.2.2

By using Euler's method of a fixed rate mortgage using an h value of 0.01 and while paying \$4500 per month it will take 22.56 years to pay off the mortgage.

3.2.3

With case one, they will pay \$924,734.57 in interest. With case two, they will pay \$468,098.10 in interest. By paying just \$500 more each month, they will save \$456,636.47!

3.2.4

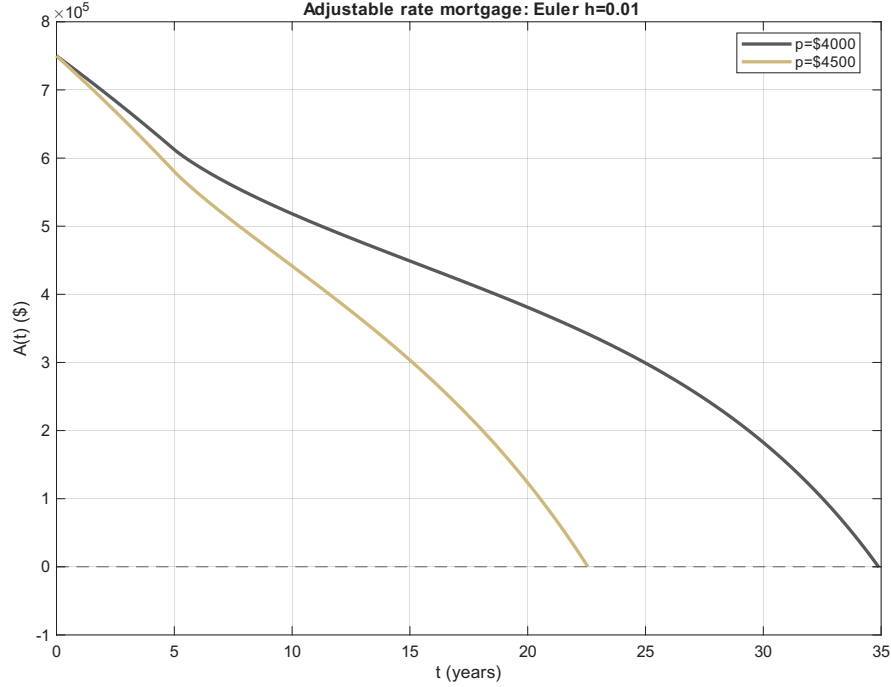


Figure 3.3: ARM (h=0.01)

The variable interest rate starts affecting the graph immediately, the total principal left on the mortgage with the ARM is immediately lower than the fixed rate. At the \$4,500 per month this trend continues for the entirety of the loan, where it is paid off sooner than the fixed rate at this payment value. At the \$4,000 per month this trend only continues until 20.5 years into the life of the mortgage. At this point in time the principal of the ARM becomes greater than the fixed rate mortgage.

4 Conclusion

When the fixed rate mortgage data is analyzed, the primary advantage is the predictability of both the loan's cost and monthly obligation being consistent over the duration of the loan. When analyzing the data, there is a big trade off between monthly affordability and total cost. While the 30 year payment plan offers a low immediate payment, the total cost of the loan due to interest is substantially higher than what it would be using an adjustable rate mortgage model.

While calculating our values, it is found that the accuracy of Euler's method depended on the chosen step size, h . When approximating the total loan duration for the mortgage, a greater h value such as $h=0.5$ will lead to a decrease in accuracy. However, by reducing the h step value to a value of $h = 0.01$, the numerical estimation of the loan duration will align much closer to the true values.

When analyzing our adjustable rate mortgage data, the main takeaway is the impact of monthly payments on the duration of the loan. Our numerical values using Euler's method demonstrated that while paying \$4000 a month, the duration of the loan equates to almost 35 years. However, increasing the monthly payment to \$4500, the duration of the loan drops to around 25 years. Paying an extra \$500 per month can save you a massive \$456,636.47 in total interest.

5 Appendix

.1 Solving Eq. (3) for A_0 , r , and p

The differential equation is:

$$\frac{dA}{dt} = rA - 12p$$

$$\frac{dA}{dt} - rA = -12p$$

Integrating Factor

$$\mu(t) = e^{-rt}$$

$$e^{-rt} \frac{dA}{dt} - re^{-rt} A = -12pe^{-rt}$$

$$\frac{d}{dt} (Ae^{-rt}) = -12pe^{-rt}$$

$$\begin{aligned} Ae^{-rt} &= \int -12pe^{-rt} dt + C \\ &= \frac{-12p}{-r} e^{-rt} + C \end{aligned}$$

$$\begin{aligned} Ae^{-rt} &= \frac{12p}{r} e^{-rt} + C \\ A &= \frac{12p}{r} + Ce^{rt} \end{aligned}$$

Apply Initial Condition $A(0) = A_0$

$$\begin{aligned} A_0 &= \frac{12p}{r} + C \\ C &= A_0 - \frac{12p}{r} \end{aligned}$$

Final Solution:

$$A(t) = \frac{12p}{r} + \left(A_0 - \frac{12p}{r} \right) e^{rt}$$

.2 MATLAB Code

```
%../APPM2360/Project01/proj1.m
s = settings;
s.matlab.appearance.figure.GraphicsTheme.TemporaryValue = light ;

function exportGraph(name, figHandle)
    plotsFolder = 'Project01/plots';

    if ~exist(plotsFolder, 'dir')
        mkdir(plotsFolder);
    end

    pdfFilename = fullfile(plotsFolder, [name, '.pdf']);
    exportgraphics(figHandle, pdfFilename, 'ContentType', 'vector');
end
```

```

function A = equation01(t, r, n, A0)
    A = A0 * (1 + r/n).^(n*t);
end

function A = equation02(t, r, A0)
    A = A0 * exp(r*t);
end

function dAdt = equation03(r, A, p, A0)
    dAdt = r*A - 12*p;
end

%#2.1
ns = [1 2 4 12]; % number of compounds per year
comp_n = zeros(size(ns));

% Total cost after 5 years
for i = 1:length(ns)
    comp_n(i) = equation01(5, 0.03, ns(i), 750000);
end

fprintf('Total cost after 5 years:\n');
for i = 1:length(ns)
    fprintf('  n=%d: $%.2f\n', ns(i), comp_n(i));
end

continuous_value = equation02(5, 0.03, 750000);
fprintf('  Continuous: $%.2f\n\n', continuous_value);

% Plot for 0 to including 30 years
t = 0:0.1:30;

f = figure;
plot(t, equation01(t, 0.03, 4, 750000), 'Color', '#CFB87C', 'LineWidth', 2);
hold on;
plot(t, equation01(t, 0.03, 12, 750000), 'Color', '#A2A4A3', 'LineWidth', 2);
plot(t, equation02(t, 0.03, 750000), 'Color', '#0A3758', 'LineWidth', 2);
xlabel('Time (years)');
ylabel('Loan Value ($)');
title('Loan Value: Compounded 4x/year, 12x/year, and Continuously');
legend('n = 4', 'n = 12', 'Continuous', 'Location', 'northwest');
grid on;
hold off;
exportGraph('3.1.1', f);

%#3.1
A0 = 750000;
r = 0.05;
p = 4000;

% (1) Euler with h = 0.5 until payoff

```

```

h = 0.5;
t = 0;
A = A0;
k = 1;
maxYears = 200;

while A(k) >= 0 && t(k) < maxYears
    t(k+1) = t(k) + h;
    A(k+1) = A(k) + h*(r*A(k) - 12*p);
    k = k + 1;
end

tpay_h05 = t(end);
fprintf('Fixed rate (r=%.2f, p=$%.0f): Euler h=%.2f payoff ~ %.2f\n', r, p, h, tpay_h05);

% (2) Plot Euler and true solution together h=0.5
A_true = (12*p)/r + (A0 - (12*p)/r).*exp(r*t);

f = figure;
plot(t, A, 'LineWidth', 2, 'Color', '#565A5C'); hold on;
plot(t, A_true, 'LineWidth', 2, 'Color', '#CFB87C');
yline(0, 'k--');
xlabel('t (years)'); ylabel('A(t) ($)');
title(sprintf('Fixed rate: Euler vs True (h = %.2f)', h));
legend('Euler', 'True', 'Location', 'best');
grid on; hold off;
exportGraph('3.2.2', f);

% (3) Repeat for h=0.01
h = 0.01;
t2 = 0;
A2 = A0;
k = 1;

while A2(k) >= 0 && t2(k) < maxYears
    t2(k+1) = t2(k) + h;
    A2(k+1) = A2(k) + h*(r*A2(k) - 12*p);
    k = k + 1;
end

tpay_h001 = t2(end);

fprintf('Fixed rate (r=%.2f, p=$%.0f): Euler h=%.2f payoff ~ %.2f\n\n', r, p, h, tpay_h001);

A_true2 = (12*p)/r + (A0 - (12*p)/r).*exp(r*t2);

f = figure;
plot(t2, A2, 'LineWidth', 2, 'Color', '#565A5C'); hold on;
plot(t2, A_true2, 'LineWidth', 2, 'Color', '#CFB87C');
yline(0, 'k--');
xlabel('t (years)'); ylabel('A(t) ($)');
title(sprintf('Fixed rate: Euler vs True (h = %.2f)', h));
legend('Euler', 'True', 'Location', 'best');
grid on; hold off;

```

```

exportGraph('3.2.3', f);

%#3.2
A0 = 750000;
h = 0.01;

% Scenario 1: p = 4000
p1 = 4000;

tA = 0;
AA = A0;
interestA = 0;

k = 1;
while AA(k) >= 0 && tA(k) < maxYears
    if tA(k) <= 5
        rk = 0.03;
    else
        rk = 0.03 + 0.015*sqrt(tA(k) - 5);
    end

    interestA = interestA + h*(rk*AA(k));
    tA(k+1) = tA(k) + h;
    AA(k+1) = AA(k) + h*(rk*AA(k) - 12*p1);
    k = k + 1;
end

tpay_p4000 = tA(end);

% Scenario 2: p = 4500
p2 = 4500;

tB = 0;
AB = A0;
interestB = 0;

k = 1;
while AB(k) >= 0 && tB(k) < maxYears
    if tB(k) <= 5
        rk = 0.03;
    else
        rk = 0.03 + 0.015*sqrt(tB(k) - 5);
    end

    interestB = interestB + h*(rk*AB(k));
    tB(k+1) = tB(k) + h;
    AB(k+1) = AB(k) + h*(rk*AB(k) - 12*p2);
    k = k + 1;
end

tpay_p4500 = tB(end);

fprintf('ARM (Euler h=%.2f):\n', h);
fprintf(' p=%d payoff ~ %.2f years, interest paid ~ %.2f\n', p1,
    tpay_p4000, interestA);
fprintf(' p=%d payoff ~ %.2f years, interest paid ~ %.2f\n', p2,

```

```

    tpay_p4500, interestB);

f = figure;
plot(tA, AA, 'LineWidth', 2, 'Color', '#565A5C');
hold on;
plot(tB, AB, 'LineWidth', 2, 'Color', '#CFB87C');
yline(0, 'k--');
xlabel('t (years)'); ylabel('A(t) ($)');
title('Adjustable rate mortgage: Euler h=0.01');
legend(sprintf('p=$%d (payoff ~ %.2f yrs)', p1, tpay_p4000), ...
        sprintf('p=$%d (payoff ~ %.2f yrs)', p2, tpay_p4500), ...
        'Location', 'best');
grid on; hold off;
exportGraph('3.2.4', f);

```